STUDY

Battery Storage Cost Curves: Market Insight

An in-depth study focusing on battery storage cost curves, addressing discrepancies in published data, benchmarking current costs, analyzing key drivers, and exploring trends and emerging technologies. This report is tailored for industry professionals and investors, using the most recent data available.

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STUDY REPORT

Battery Storage Cost Curves: Market Insight

Executive Summary

Finding 1 — Battery storage becomes a core flexibility asset

By 2025-2026, battery storage is expected to move beyond a niche balancing tool and become a primary asset for energy flexibility. This transition is anchored by the broad adoption of Lithium Iron Phosphate (LFP) in stationary applications because of its cost-effectiveness and durability.

Finding 2 — LFP standardizes the market, with alternatives emerging

LFP is projected to reach over 90% installation share by 2025, making it the dominant standard in stationary storage. At the same time, sodium-ion is gaining traction as a way to reduce raw material risk and cost volatility, while solid-state batteries retain longer-term potential but are not yet cost-competitive.

Finding 3 — No single global battery cost curve applies

Forecasts indicate lithium-ion prices could still average above $100/kWh globally in 2025-2026, but regional deviations are material. China is reducing costs aggressively through scale and supply chain efficiency, while Europe and the US face structurally higher costs, undermining the usefulness of a single global average.

Finding 4 — Methodology materially alters cost comparisons

Divergent published cost curves reflect different assumptions, boundaries, and scenario definitions rather than simple market disagreement. Decision-makers need to normalize parameters across cell, pack, and fully installed system costs and separate endogenous learning effects from broader scenario overlays to make analyses decision-useful.

Finding 5 — Policy and non-cell costs are major price drivers

Trade measures, tariffs, material volatility, and non-cell hardware costs are significant determinants of delivered battery system economics. These effects are especially pronounced in import-heavy markets such as the US, and balance-of-system costs are not falling as quickly as cell prices, complicating headline cost declines.

Finding 6 — Market growth is set to accelerate sharply

Global battery storage deployment could exceed 100 GW annually by 2025, led by the US and China. Growth is being driven by utility-scale buildouts and integration with renewable energy projects, reinforcing storage's expanding role in power systems.

1

Introduction and Context

1

Introduction and Context

Battery storage entered 2025 and 2026 as a core flexibility asset rather than a niche balancing technology. Global deployment set new records, LFP became the dominant chemistry in stationary applications, and pricing continued to fall, but the market also became more segmented by geography, policy regime, duration requirement, and cost accounting boundary, which is precisely why published cost curves now diverge more visibly than headline averages suggest.[19] [20] [22] [21]

For investors, planners, and policymakers, the practical issue is not whether battery costs are falling, but which cost curve is decision-relevant for a given use case. A cell, pack, containerized system, and fully installed four hour project can all be described as "battery cost," yet they answer different questions about manufacturing competitiveness, procurement timing, project finance, resource planning, and policy design.[21] [28] [3]

1.1

State of the Global Battery Storage Market in 2025-2026

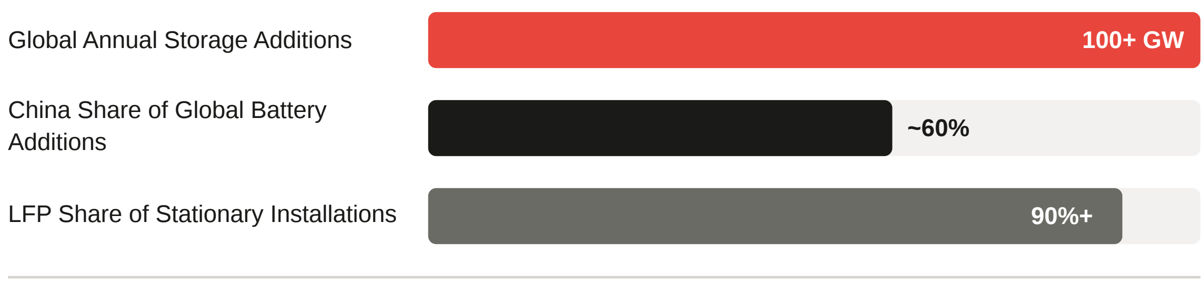
- Global battery storage moved into a new scale phase in 2025. BloombergNEF reported that annual global energy storage additions exceeded 100 GW for the first time in 2025, while the IEA reported that China accounted for around 60% of global battery storage additions in 2025, followed by the United States and Europe.[22] [19]
- Chemistry and application trends became more concentrated. The IEA states that LFP accounted for over 90% of global stationary battery storage installations in 2025, reflecting its cost and cycle life advantages, while project duration continued to lengthen as markets shifted from fast response services toward energy shifting and renewable integration.[20] [28]
- Cost declines resumed sharply, but not uniformly across segments or regions. BloombergNEF's 2025 survey put the global average lithium ion pack price at $108/kWh, China at $84/kWh, and North America and Europe at 44% and 56% above China respectively; separately, stationary storage pack prices were reported at $70/kWh in 2025, showing how segment mix can materially change the benchmark cited.[21] [23] [30]

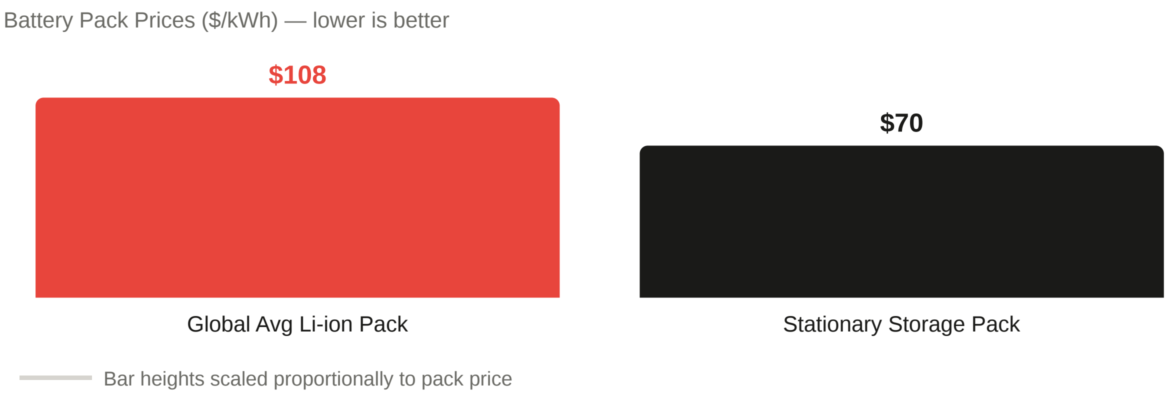
Title — Selected 2025 battery storage market signals

INDICATOR	2025 VALUE	MARKET IMPLICATION
Global annual storage additions	Over 100 GW	Storage reached utility scale mainstream adoption.
China share of global battery additions	Around 60%	Supply chain and deployment leadership remained concentrated.
LFP share of stationary installations	Over 90%	Low cost chemistry became the market standard.
Global average lithium ion pack price	$108/kWh	Headline battery prices continued to decline.

INDICATOR	2025 VALUE	MARKET IMPLICATION
Stationary storage pack price	$70/kWh	Storage specific pack costs fell below broad market averages.

Source: [19] [20] [22] [21] [30]





1.2

Why Cost Curve Disagreements Matter

- Cost curve disagreements change asset valuation. A forecast built on cell or pack prices can materially understate the capital required for a delivered project once inverters, thermal management, controls, EPC, interconnection, land, and soft costs are included, especially in higher cost regions.[21] [28]
- Cost curve disagreements change planning outcomes. Different assumptions on duration, degradation, learning rates, and service value can alter conclusions on whether storage is best suited for ancillary services, renewable firming, capacity adequacy, or long duration substitution, which directly affects resource plans and market design.[2] [12] [3]
- Cost curve disagreements change policy interpretation. Regional price gaps driven by tariffs, local content rules, imported battery premiums, and manufacturing overcapacity mean that a global average can be directionally useful but operationally misleading for domestic industrial policy or procurement design.[23] [36] [19]

In practice, disagreements are not merely analytical noise. They can shift hurdle rates, merchant revenue expectations, procurement timing, and the apparent competitiveness of storage against gas peakers, transmission upgrades, or other flexibility options.[25] [27]

For this reason, the central discipline in battery market analysis is matching the curve to the decision. Investors need financeable installed cost and revenue durability, planners need technology consistent system cost trajectories, and policymakers need region specific cost evidence that reflects trade and industrial policy realities rather than global manufacturing averages.[28] [21] [11]

1.3
Scope and Structure of This Report

This report is designed as a decision support study for investors, developers, utilities, planners, and policymakers evaluating battery storage cost trajectories in 2025 and 2026. It focuses on why published cost curves disagree, what current benchmarks actually mean, and how to select the right source and cost boundary for a specific analytical task.[19] [21]

- Cost curve foundations: The report first separates cell, pack, and full installed system definitions, then compares bottom up, top down, and hybrid methodologies, including learning rate and scenario assumptions that drive divergence across published curves.[3] [11]
- Market benchmarks and drivers: It then reviews current 2025 to 2026 cost benchmarks by region and chemistry, with emphasis on LFP dominance, China versus Europe versus United States pricing gaps, and the role of tariffs, supply chains, balance of system costs, and commodity volatility.[23] [20] [36]
- Source comparison and outlook: The final sections compare major market and modeling sources, assess deployment outlooks including the move into the 100 GW era globally and continued U.S. expansion, and examine emerging technologies such as sodium ion, solid state, flow batteries, and gravity based storage to frame long term cost optionality.[22] [25] [11]

The report's organizing principle is practical comparability. Each later section is intended to help readers translate headline battery price claims into investment grade, planning grade, or policy grade interpretations, reducing the risk of using an accurate number in the wrong analytical context.[2] [27]

2

Why Published Cost Curves Disagree

2

Why Published Cost Curves Disagree

Published battery storage cost curves disagree primarily because they often measure different things, use different construction methods, and embed different forward assumptions. A curve built from cell manufacturing economics is not directly comparable with one built from delivered pack transactions or with one built from a fully installed four hour grid project that includes power conversion, thermal systems, controls, EPC, interconnection, and developer overhead.[3] [50] [58]

Disagreement also widens when analysts mix empirical market observations with engineering models and then apply different learning rates, duration assumptions, utilization profiles, and policy scenarios. For investors and industry professionals, the practical implication is that divergence across published curves is usually methodological rather than erroneous, and the correct response is to normalize boundaries, assumptions, and scenario definitions before drawing valuation or strategy conclusions.[2] [48] [27]

2.1

Cell vs. Pack vs. Full Installed System Cost Definitions

The most common source of disagreement is cost boundary definition. Cell cost usually refers to electrochemical cells at the manufacturing level, pack cost adds module or pack integration and battery management components, and full installed system cost extends further to include inverter or PCS, containerization, thermal management, fire protection, controls, installation labor, EPC margin, site work, and other balance of system items.[3] [50] [61]

- Cell cost: Most useful for assessing manufacturing competitiveness, chemistry economics, and learning at the electrochemical level, but it excludes much of what project developers actually buy and install.[3]
- Pack cost: Closer to a procurement benchmark and widely cited in market commentary, but still incomplete for project finance because it omits substantial non-battery hardware and soft costs.[20] [50]
- Full installed system cost: Most decision relevant for utility and infrastructure investors because it captures the delivered asset boundary, including BOS and EPC elements that can materially change regional economics.[58] [61]

Title — Cost boundary differences in published curves

COST BOUNDARY	TYPICALLY INCLUDES	WHY CURVES DIVERGE
Cell	Electrochemical cell manufacturing cost or price.	Lowest boundary and most sensitive to chemistry and factory scale.
Pack	Cells plus integration, enclosure, wiring, and BMS.	Often used in market surveys, but still excludes project delivery costs.
System equipment	Pack plus inverter or PCS, thermal and safety systems, container or cabinet.	Captures storage equipment sold to developers, but not full site execution.

COST BOUNDARY	TYPICALLY INCLUDES	WHY CURVES DIVERGE
Full installed project	System equipment plus BOS, labor, EPC, permitting, interconnection, and overhead.	Highest and most finance relevant boundary, with strong regional variation.

Source: [50 61 58]

A second definitional problem is the unit of expression. Some sources report $/kWh, others $/kW, and duration changes the relationship between the two, so a two hour and four hour system can appear cheaper or more expensive depending on which denominator is used.[50 53]

Published curves also differ on whether they represent nameplate energy, usable energy, beginning of life capacity, or augmented lifetime capacity. Those choices affect apparent cost even before any forecasting begins, especially when degradation reserves or augmentation strategies are embedded in the system design.[2 41]

2.2

Bottom-up vs. Top-down vs. Hybrid Methodologies

Methodology is the second major reason curves disagree. Bottom-up studies build cost from components and engineering assumptions, top-down studies infer cost from observed prices or market transactions, and hybrid approaches anchor engineering models to real market data to reduce the weaknesses of either method alone.[50 48 3]

- Bottom-up: Strong for transparency because each component can be inspected, stress tested, and localized by region, but results depend heavily on assumed bill of materials, labor rates, margins, and system architecture.[50 58]
- Top-down: Strong for capturing what buyers actually paid or what suppliers actually quoted, but observed prices can reflect temporary oversupply, strategic discounting, bundled contracts, or opaque scope definitions.[20 27]
- Hybrid: Often the most decision-useful because it reconciles engineering structure with market evidence, but disagreement remains if the calibration year, geography, or procurement channel differs across studies.[48 50]

Bottom-up models tend to produce smoother and more internally consistent cost curves because they are built from assumed component trajectories. That makes them useful for long-range planning, but they can lag fast market moves when cell oversupply, freight swings, or tariff changes cause transaction prices to move faster than engineering models are updated.[48 27]

Top-down curves can move quickly with the market, but they may compress unlike products into a single average. A survey based on broad lithium-ion pack prices, for example, can blend EV, stationary, LFP, and NMC segments, while a utility-scale storage benchmark may isolate a specific stationary architecture and therefore show a very different trajectory.[20 3]

A further methodological split concerns whether studies model only capex or full life cycle economics. Analyses that include degradation, augmentation, round-trip efficiency, replacement timing, and operating strategy can reach different effective cost conclusions even when starting from similar upfront capex assumptions.[2 41 42]

2.3

Learning Rate Assumptions and Scenario Definitions

Even when two studies start from similar current costs, they can diverge sharply because they assume different future learning rates and different deployment scenarios. Learning curves link cost decline to cumulative production or deployment, so disagreement over future manufacturing scale, chemistry mix, and performance improvement translates directly into disagreement over projected cost.[3, 48]

- Learning rate choice: Some studies estimate decline from historical price series alone, while others adjust for energy density, performance, or service delivered, which can produce materially different implied improvement rates.[3]
- Scenario framing: Conservative, moderate, and advanced scenarios often differ not only in cost decline speed but also in assumptions about manufacturing utilization, policy support, raw material availability, and technology substitution.[48, 27]
- Duration and use case: A short-duration ancillary services battery and a four to eight hour renewable shifting asset do not share the same cost structure, augmentation profile, or revenue logic, so scenario definitions that ignore duration can create false comparability.[50, 2]

Scenario definitions also differ on exogenous factors that are not pure learning effects. Tariffs, local content rules, interconnection delays, financing conditions, and regional labor costs can flatten or steepen realized cost curves even if underlying battery manufacturing costs continue to fall, which is why global learning curves often diverge from regional installed cost curves.[27, 58]

Another source of disagreement is whether the curve assumes chemistry continuity or chemistry transition. Forecasts that assume continued LFP dominance, for example, will differ from those that assume faster penetration of sodium-ion or other alternatives, because the cost path depends on both learning within a chemistry and substitution across chemistries.[20]

For analysts, the key discipline is to separate endogenous learning from scenario overlays. If one source embeds aggressive deployment, benign commodity prices, stable trade policy, and longer-duration standardization, while another assumes slower buildout and higher regional frictions, the resulting disagreement is expected and should be interpreted as scenario variance rather than a contradiction in the underlying economics.[48, 27]

3

Current Cost Benchmarks by Region and Technology

3

Current Cost Benchmarks by Region and Technology

Current battery cost benchmarks are converging downward, but the market is no longer well described by a single global average. The most decision-relevant picture for 2025 to 2026 is a three-layer one: a global lithium-ion pack average still above $100/kWh, a stationary-storage segment already materially below that level, and a China-led low-cost frontier that is pulling the industry toward the $80/kWh threshold faster than Europe and the United States can currently match.[21] [20]

Technology mix is central to interpreting these benchmarks. LFP has become the default chemistry for stationary storage because its lower materials cost, strong cycle life, and reduced dependence on nickel and cobalt align with grid applications better than higher-energy chemistries, while sodium-ion is beginning to move from pilot status into early commercial deployment and solid-state remains strategically important but not yet a near-term cost benchmark for mainstream stationary systems.[20] [19] [93]

3.1

Global Pack Cost Trends: The Race to $80/kWh and Below

The global volume-weighted average lithium-ion battery pack price fell to $108/kWh in 2025, down 8% from 2024, according to BloombergNEF. That headline average matters, but it masks a much faster decline in stationary storage, where pack prices reportedly fell to $70/kWh in 2025, making storage the lowest-priced battery segment for the first time in BNEF's survey.[21]

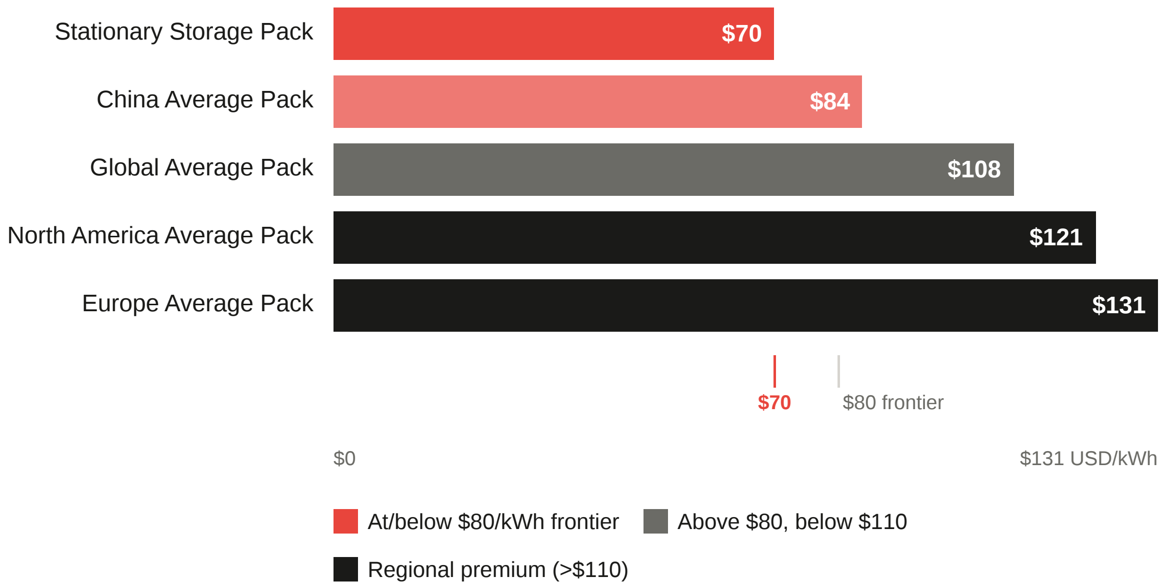
For investors, the practical implication is that the race to $80/kWh is no longer a theoretical industry milestone. It has effectively been reached or surpassed in parts of the stationary segment and in the Chinese market, while the broader global average remains above that level because it still blends higher-cost automotive and non-China procurement channels.[21] [20]

- Segment effect: The gap between $108/kWh global average packs and $70/kWh stationary packs shows why cross-sector averages should not be used as direct proxies for grid storage procurement.[21]
- Cost-down mechanism: BloombergNEF attributes the 2025 decline primarily to cell manufacturing overcapacity, intense supplier competition, and the continued shift toward lower-cost LFP chemistries.[21]
- Interpretation caution: The approach to $80/kWh at pack level does not mean fully installed project costs are near $80/kWh, because PCS, thermal systems, fire protection, controls, EPC, and interconnection remain outside the pack boundary.[20] [3]
- Trend direction: Historical learning evidence suggests stationary applications can continue to decline rapidly because they face fewer energy-density constraints than EV batteries, supporting the view that sub-$80/kWh pack pricing can broaden if oversupply and LFP scale persist.[3]

Title — 2025 battery pack benchmarks

BENCHMARK	2025 VALUE (USD/KWH)	MARKET READING
Global average lithium-ion pack	108	Broad cross-sector average still above $100/kWh.
China average pack	84	Closest major-market benchmark to the $80/kWh frontier.
North America implied average pack	121	About 44% above China, showing regional premium.
Europe implied average pack	131	About 56% above China, the highest-cost major region in BNEF data.
Stationary storage pack	70	Storage-specific segment already below the $80/kWh threshold.

Source: [21]



3.2

Regional Divergence: China, Europe, and the United States

Regional divergence is now one of the most important features of battery cost benchmarking. BloombergNEF's 2025 survey places China at $84/kWh on average, while North America and Europe are 44% and 56% higher respectively, implying roughly $121/kWh for North America and $131/kWh for Europe.[21]

China's lead reflects a combination of manufacturing scale, dense supply-chain localization, aggressive competition, and a faster shift into low-cost LFP-heavy product mixes. The IEA also reports that China accounted for around 60% of global battery storage additions in 2025, reinforcing the feedback loop between domestic deployment scale and manufacturing cost leadership.[19] [21]

Europe and the United States face structurally higher costs even as global cell prices fall. The main causes are higher local production costs, greater dependence on imported batteries or imported upstream materials, and policy-driven frictions that can preserve domestic manufacturing optionality but widen near-term procurement premiums relative to China.[21]

- China: The lowest-cost major market in 2025, already near the $80/kWh benchmark, with cost leadership reinforced by scale and local supply-chain integration.[21, 19]
- United States: Higher than China by roughly 44% on BNEF's 2025 benchmark, reflecting a mix of domestic cost structure, imported battery premiums, and a market that still carries more non-China sourcing friction.[21]
- Europe: The highest-cost region among the three at about 56% above China, despite benefiting from Chinese export competition in some channels.[21]
- Analytical implication: A global average is useful for macro trend direction, but regional procurement, project finance, and policy analysis should use localized benchmarks because the spread between China and Europe is now too large to treat as noise.[21, 20]

3.3

LFP Dominance and Emerging Chemistries

LFP is now the defining chemistry of the stationary storage market. The IEA reports that LFP accounted for over 90% of global stationary battery storage installations in 2025, and that LFP battery packs were more than 40% cheaper on average than NMC alternatives per kWh in 2025, which explains why the chemistry has become the benchmark for mainstream grid storage cost curves.[20, 19]

The dominance of LFP is not only a price story. It also reflects fit-for-purpose economics: stationary systems value cycle life, thermal stability, and lower raw-material exposure more than maximum energy density, which weakens the case for nickel-rich chemistries in most four-hour and shorter grid applications.[19, 2]

Sodium-ion is the most credible near-term challenger in low-cost stationary storage, but it remains an emerging rather than dominant chemistry as of August 25, 2026. The IEA says CATL confirmed commercial-scale sodium-ion deployment beginning in 2026, and CATL announced a three-year 60 GWh sodium-ion storage agreement with HyperStrong and described sodium storage as entering the GWh-scale deployment era, which is commercially significant even if it does not yet displace LFP's incumbent position.[98, 94, 93]

Solid-state batteries remain strategically important, but they are not yet a current cost benchmark for mainstream stationary storage. Available evidence points to ongoing development and evaluation activity rather than broad, bankable grid-scale commercialization, so they should be treated as a longer-dated option rather than part of the present benchmark set.[106, 20]

- LFP cost position: More than 40% cheaper on average than NMC packs per kWh in 2025, reinforcing its role as the default chemistry for cost-sensitive storage deployments.[20]
- LFP market position: Over 90% share of global stationary installations in 2025 indicates that alternative lithium chemistries are now niche in mainstream stationary applications.[20]
- Sodium-ion outlook: Scenario work suggests sodium-ion could reach competitiveness with low-cost lithium-ion variants in the early 2030s under favorable learning and supply-chain conditions, but that is still a forward-looking pathway rather than a present market equilibrium.[11]

- Market conclusion: For current benchmarking, investors should treat LFP as the base case, sodium-ion as an emerging optionality with early commercial traction, and solid-state as a longer-horizon technology signal rather than a 2025 to 2026 cost anchor.[20, 93, 106]

4

Key Cost Drivers and Risk Factors

4

Key Cost Drivers and Risk Factors

Battery storage cost curves are increasingly shaped by factors outside the cell factory gate. In 2025 and 2026, trade measures, supply chain concentration, non-cell hardware, and renewed volatility in battery materials all affected delivered system costs, often in ways that widened the gap between falling global pack benchmarks and the costs actually faced by developers and investors in specific markets.[120, 123, 50]

For analytical purposes, the key point is that these drivers do not act symmetrically. Trade and logistics frictions are highly regional, BOS and soft costs are sticky relative to cell prices, and commodity shocks affect chemistries differently, especially in a market now dominated by LFP for stationary storage.[20, 125, 156]

4.1

Tariffs, Trade Policy, and Supply Chain Constraints

Trade policy became a first-order cost variable in 2025 and 2026, particularly for import-dependent markets. The IEA reports that higher tariffs and duties in the United States increased domestic production’s share of demand for battery cells by 25 percentage points during 2023 to 2025, but also raised average delivered costs, with production tax credits and shifting trade patterns offsetting only around half of the increase across imports and domestic production.[120]

Supply chain resilience policy is now interacting directly with project capex. The IEA states that 32 countries had policies in place by April 2026 to incentivize supply chain resilience and diversification across critical minerals and clean energy technologies, while 45 new policies affecting trade in key clean energy technologies were implemented in 2025 in addition to broad US tariff measures.[121]

- Cost transmission mechanism: Tariffs affect storage costs not only through imported cells and packs, but also through cathode materials, graphite anodes, power electronics, and manufacturing equipment, so the delivered cost impact is broader than the headline duty on finished batteries.[123, 82]
- Concentration risk: China remains the dominant supplier across multiple battery chain segments, with the IEA noting in 2025 data that China supplied almost 85% of cathode active materials and over 90% of anode active material production, which leaves non-China buyers exposed to policy and logistics shocks even when cell prices are falling.[82]
- Export-control risk: The IEA reports that China announced export controls in October 2025 on cathode materials, cathode precursors, graphite anode materials, and battery manufacturing equipment and technologies, highlighting how upstream chokepoints can rapidly alter procurement risk and lead times.[123]
- Project-level implication: Wood Mackenzie estimates that, under higher-tariff scenarios, utility-scale storage projects in the United States could see cost increases ranging from 12% to more than 50%, reflecting the sector’s heavy dependence on Chinese battery cells.[125]
- Near-term market evidence: S&P Global reported on August 24, 2026 that US lithium-ion battery imports rebounded in the second quarter of 2026 to 215,942 metric tons, up 26% quarter on

quarter but down 10% year on year, indicating that trade friction is reshaping sourcing patterns rather than eliminating import dependence.[127]

For investors, the main conclusion is that trade policy now changes the slope of regional cost curves, not just the level. A global pack benchmark may still fall, but the realized installed-cost curve in a protected or supply-constrained market can flatten temporarily if tariffs, local-content rules, or export controls outpace manufacturing localization.[120] [36]

4.2

Balance-of-System, Soft Costs, and Non-Cell Hardware

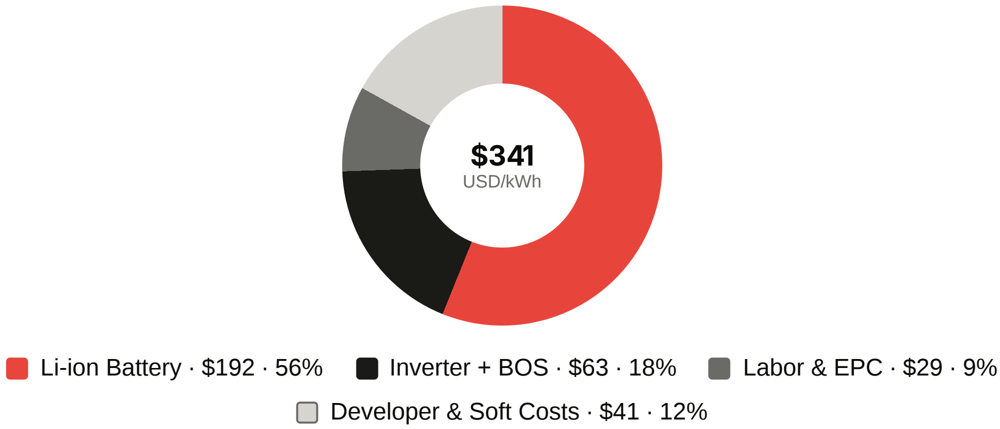
As cell and pack prices decline, non-cell costs become more visible in total project economics. NREL's ATB framework continues to model utility-scale battery systems as the sum of battery pack cost plus BOS cost, and explicitly notes that battery components are projected to decline faster than inverters, BOS, installation, and soft-cost components.[50] [136]

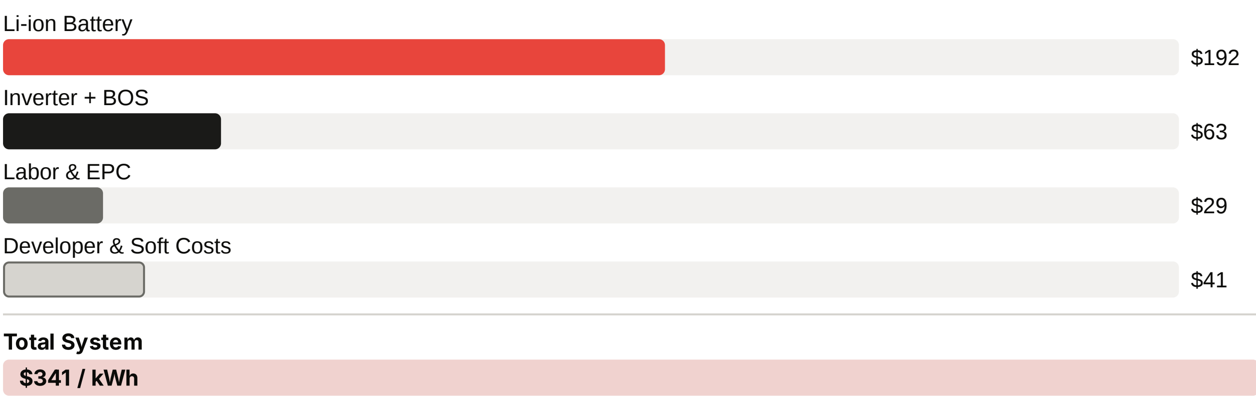
This creates a structural floor under installed system costs. Even if pack prices approach or move below the $80/kWh frontier in leading markets, developers still pay for PCS or inverter equipment, thermal management, fire protection, structural and electrical BOS, installation labor, EPC overhead, permitting, interconnection, contingency, and developer margin.[51] [50]

Title — Illustrative utility-scale BESS cost stack

COST COMPONENT	4 HOUR COST (USD/KWH)	COST IMPLICATION
Lithium ion battery	192	Largest single component, but not a majority of full system cost.
Inverter plus structural and electrical BOS	63	Core non-cell hardware that scales with power architecture and site design.
Installation labor and EPC overhead	29	Sticky execution costs that do not fall as fast as cells.
Developer cost, interconnection, contingency, profit	41	Soft-cost layer that materially separates equipment cost from financeable capex.
Total energy storage system cost	341	Demonstrates why pack benchmarks understate delivered project cost.

Source: [51]





- Cost-curve implication: When battery pack prices fall faster than PCS, BOS, and soft costs, the percentage share of non-cell items rises, so total installed cost declines more slowly than pack cost alone.[50]
- Duration effect: BOS and power-related costs are partly fixed on a $/kW basis, which is why longer-duration systems can look cheaper in $/kWh terms while still requiring substantial non-energy hardware and site work.[135]
- Regional effect: Labor, permitting, tax, and interconnection costs vary far more by geography than cell manufacturing costs, making BOS and soft costs a major reason regional installed-cost curves diverge even when pack procurement is global. This is an inference supported by NREL's bottom-up structure and the IEA's evidence of regional cost divergence.[51] [156]
- Forecasting implication: Analysts should expect diminishing pass-through from future cell price declines into full project capex unless inverter, thermal, fire-safety, and interconnection costs also compress.[141]

For project finance, this means the relevant risk is not only whether batteries get cheaper, but whether the non-cell share becomes the dominant source of variance. In mature markets, that variance increasingly comes from site-specific execution, grid-connection complexity, and procurement of power electronics rather than from the electrochemical stack alone.[51] [27]

4.3

Commodity Price Volatility (Lithium, Cobalt, Nickel)

Battery material volatility re-emerged as a material risk in late 2025 and 2026 after the price correction of 2023 and 2024. The IEA reports that lithium prices more than doubled between early 2025 and early 2026, though they remained around 70% below the 2022 peak, while cobalt prices rose by around 130% due largely to export restrictions in the Democratic Republic of the Congo.[123] [20]

The cost pass-through from these metals is chemistry-specific. The IEA notes that cathode active material is the single most important component in cell production cost, accounting for 40% to 50% for NMC batteries and 25% to 30% for LFP batteries, which helps explain why LFP-dominant stationary storage is less exposed to cobalt and nickel shocks than nickel-rich EV chemistries.[20]

- Lithium: The most important raw-material risk for stationary storage because LFP still depends on lithium even while avoiding nickel and cobalt. The IEA attributes the 2025 to 2026 rebound to faster-than-anticipated demand growth, especially from battery energy storage, low inventories in China, and temporary supply disruptions.[20]
- Cobalt: More relevant to NMC exposure than to mainstream stationary LFP systems, but still important for diversified battery portfolios. The IEA states that the DRC accounts for almost two-thirds of global cobalt supply, so export bans and quotas can move prices quickly.[20]
- Nickel: Less central to LFP-heavy stationary storage, but still relevant for projects using nickel-bearing chemistries and for broader battery-sector investment cycles. The IEA notes that Indonesia and China together accounted for over three-quarters of total supply growth across key energy minerals between 2023 and 2025, underscoring concentration risk even in periods of oversupply.[150]
- Volatility profile: The IEA's 2025 outlook states that three-quarters of critical minerals showed greater price volatility than oil, and half were more volatile than natural gas, which is a useful benchmark for stress-testing storage cost assumptions.[153]
- System-level sensitivity: The IEA estimates that a tripling of battery material prices would raise the final price of EVs and storage systems by around 5%, which indicates that materials matter, but also that today's battery cost structure is more buffered than in earlier years because of chemistry shifts, scale, and non-material cost shares.[123]

Recent academic work reinforces the point that energy-transition metals exhibit heterogeneous and difficult-to-predict volatility patterns rather than a single common cycle, which supports using scenario ranges rather than point estimates for lithium, cobalt, and nickel in cost-curve modeling.[116]

For stationary storage, the practical conclusion is that lithium remains the most direct commodity risk, while cobalt and nickel increasingly matter through portfolio mix, alternative chemistry exposure, and investor sentiment toward the broader battery supply chain. That is one reason LFP has improved cost resilience, but it has not eliminated raw-material risk from storage cost curves.[20] [123]

5

Major Source Comparison: Which Curve to Use and When

5

Major Source Comparison: Which Curve to Use and When

Published battery storage curves are most useful when treated as purpose-built tools rather than competing answers. The practical distinction is between sources optimized for transparent engineering normalization, sources optimized for current market price discovery, and sources optimized for commercial outlooks that combine cost, deployment, and revenue assumptions.[183, 21, 172, 174]

For investors and industry professionals, the key question is not which source is universally best, but which source matches the decision boundary. NREL is strongest for transparent installed-cost structure and scenario consistency, BloombergNEF for global pack and segment pricing, Wood Mackenzie for near-term commercial market intelligence, and S&P Global for market-linked economics and revenue-aware outlooks.[183, 21, 172, 177]

5.1

NREL ATB vs. BloombergNEF vs. Wood Mackenzie vs. S&P Global

These four sources differ less in analytical quality than in scope, boundary, and update logic. NREL's ATB and its 2025 utility-scale battery update are engineering-led projection frameworks; BloombergNEF is primarily a market-price and segment-tracking source; Wood Mackenzie is a commercial market monitor with deployment, pricing, and policy coverage; and S&P Global increasingly frames storage through market economics, levelized cost, and merchant revenue conditions.[183, 21, 172, 174]

Title — Major source comparison

SOURCE	PRIMARY LENS	MAIN STRENGTH	MAIN LIMITATION
NREL ATB and 2025 update	Bottom-up installed cost projection.	Transparent assumptions, duration treatment, and scenario consistency.	Can lag fast transaction-price moves and is less suited to live procurement.
BloombergNEF	Empirical market pricing and segment benchmarking.	Best for global pack trends, regional spreads, and chemistry segmentation.	Pack-level data can be misused as project capex if boundary is not adjusted.
Wood Mackenzie	Commercial market intelligence and quarterly monitoring.	Strong on near-term deployment, policy effects, and developer-facing market context.	Coverage is often market-specific and less transparent on full model internals.
S&P Global	Market economics, levelized cost, and revenue-aware outlooks.	Useful for linking capex trends to merchant value, spreads, and investment cases.	Less suitable as a pure equipment-cost benchmark without supplementary cost normalization.

Source: [183]

- NREL ATB: Best when the analyst needs a normalized installed-cost curve by duration and a transparent decomposition of battery pack versus BOS. The ATB framework explicitly treats utility-scale storage across multiple durations and is designed for scenario comparability across technologies, which is valuable for planning and research.[183]

- BloombergNEF: Best when the analyst needs current market evidence on pack prices, chemistry mix, and regional divergence. Its 2025 survey reported a global average lithium-ion pack price of $108/kWh and stationary storage pack prices of $70/kWh, making it highly useful for identifying where the market frontier sits, but not for inferring turnkey project capex without adders.[21]
- Wood Mackenzie: Best when the analyst needs a commercial read on what is happening in the market now, especially in the United States. Its quarterly monitor explicitly compiles deployments, prices, policies, regulations, and business models, which makes it strong for near-term market timing and policy-sensitive procurement analysis.[172]
- S&P Global: Best when the analyst needs storage economics tied to power-market outcomes rather than hardware cost alone. S&P Global's recent storage analysis emphasizes declining costs alongside falling merchant revenues in some markets, and its 2025 PJM reference deck presents battery storage levelized cost outlooks through 2045, which is useful for investment cases where revenue durability matters as much as capex.[174] [177]
- Comparative weakness across all four: None should be used unadjusted outside its intended boundary. NREL can understate short-term market dislocations, BloombergNEF can understate delivered project cost if pack data are overextended, Wood Mackenzie can be less transparent than public engineering models, and S&P Global can blend cost and market-value assumptions in ways that are not ideal for pure capex benchmarking.[183] [21] [172] [174]

5.2

When to Use Modelled Projections vs. Empirical Market Data

Modeled projections are preferable when the decision requires internal consistency across time, technologies, and scenarios. They are especially useful for integrated resource planning, policy design, and long-horizon research because they can hold duration, degradation treatment, and cost boundaries constant while testing alternative assumptions.[183] [3]

Empirical market data are preferable when the decision is procurement-sensitive, timing-sensitive, or exposed to short-term dislocations. In 2025, for example, BloombergNEF's observed pack-price declines reflected oversupply, intense competition, and segment-specific pricing that a slower-moving engineering model may not capture immediately.[21]

- Use modeled projections when: The task is to compare storage against other technologies on a like-for-like basis, estimate long-run cost trajectories, or test policy scenarios. This is the right choice for capacity-expansion models, decarbonization pathways, and academic research where methodological transparency matters more than the latest spot movement.[183] [3]
- Use empirical market data when: The task is to assess current procurement windows, supplier competitiveness, regional sourcing spreads, or whether a quoted project budget is realistic today. This is the right choice for term-sheet negotiation, vendor screening, and near-term investment committee work.[21] [172]
- Use both when: The decision spans both current execution and future strategy. A robust approach is to anchor the base year with empirical data, then extend forward with a modeled curve that explicitly states learning, duration, and policy assumptions.[183] [172]

- Key caution: Empirical data can be temporarily distorted by oversupply, discounting, or bundled contracts, while modeled data can smooth away real market shocks. The 2025 to 2026 environment, shaped by tariffs, regional divergence, and changing merchant spreads, is precisely the kind of market where relying on only one of the two approaches creates avoidable error.[21 174 172]
- Practical rule: If the question includes the words today, quote, bid, tariff, or procurement, start with empirical data. If it includes 2030, scenario, pathway, or cross-technology comparison, start with modeled projections and then sanity-check against market evidence.[183 21]

5.3

Matching the Source to the Use Case (Project Finance, Policy, Research)

Source selection should follow the decision boundary, not brand familiarity. Project finance needs financeable installed-cost and revenue realism, policy work needs transparent and scenario-consistent assumptions, and research needs reproducibility plus clearly defined cost boundaries.[183 174]

- Project finance: Start with empirical commercial sources such as Wood Mackenzie and BloombergNEF for current pricing direction, then translate those inputs into full installed project economics using an engineering framework such as NREL. Add S&P Global where merchant exposure, capture-price risk, or ancillary-service saturation materially affects debt sizing or equity returns.[172 21 183 174]
- Policy development: Use NREL first because transparent assumptions and duration-specific installed-cost treatment are more defensible in public analysis. Supplement with BloombergNEF or IEA-style market evidence when the policy question depends on current global manufacturing conditions or regional price divergence rather than only modeled long-run learning.[183 21]
- Research and academic analysis: Use NREL-type bottom-up projections and peer-reviewed learning-curve literature as the core analytical spine, because they are easier to document and replicate. Empirical sources such as BloombergNEF remain valuable as calibration points, but proprietary market averages should not be treated as self-explanatory without boundary normalization.[183 3 21]
- Corporate strategy and market entry: Use Wood Mackenzie and S&P Global when the question is where to deploy capital, which market design is improving, or how revenue pools are evolving. These sources are stronger than pure cost models at connecting storage economics to actual market structure and competitive behavior.[172 174]
- Recommended workflow: Use BloombergNEF to identify the market price frontier, NREL to convert that frontier into normalized installed-cost scenarios, Wood Mackenzie to test whether the target market is behaving in line with those assumptions, and S&P Global to assess whether falling capex is actually translating into durable project value. That combination is usually more decision-useful than relying on any single curve in isolation.[21 183 172 177]

6

Deployment Trends and Market Outlook

6

Deployment Trends and Market Outlook

Battery storage has moved from a cost-reduction story to a scale-deployment story. By 2025, the market had entered a new buildout phase in which annual additions exceeded 100 GW globally, utility-scale systems accounted for most new capacity, and deployment broadened beyond the early leader markets even as China and the United States remained dominant.[19] [22] [28]

For investors, the key implication is that battery storage is now in a supercycle defined by simultaneous demand pull from renewables integration, grid reliability, peak management, and emerging large-load growth. The outlook is therefore not just for more deployments, but for a wider geographic spread, deeper project pipelines, and a stronger linkage between storage buildout and broader power-system reconfiguration through 2030.[195] [205] [27]

6.1

US Utility-Scale Growth: 43.6 GW (End-2025) to 100+ GW by 2030

The U.S. utility-scale market is on a steep expansion path from an operational base of 43.6 GW at end-2025, after averaging roughly 70% annual growth over the prior three years. EIA also projected in early 2025 that utility-scale battery capacity would grow by 14 GW in 2025 and 11 GW in 2026, confirming that the market entered 2026 with strong near-term momentum rather than a one-off spike.[195] [201]

Crossing 100 GW by 2030 is increasingly plausible when current operating capacity is combined with the pace of annual additions now being observed and forecast. Wood Mackenzie's mid-2026 outlook projects the U.S. market to reach 200 GW of cumulative installed energy storage capacity by 2031, with 85% of 2026-2031 installations coming from the utility sector, which implies a utility-scale fleet comfortably above 100 GW around 2030 under central growth assumptions.[205]

- Growth engine: Solar-plus-storage buildout remains the primary structural driver, as storage is increasingly paired with utility-scale solar to shift midday generation into evening peaks and improve capacity value.[202] [211]
- Reliability driver: Grid operators and utilities are procuring batteries not only for arbitrage, but for resource adequacy, congestion relief, and fast-response balancing, especially in systems with rising renewable penetration.[27] [28]
- Load-growth driver: Data centers and other large loads are strengthening the case for storage as a flexible capacity resource, particularly where interconnection and thermal generation lead times are long.[204] [205]
- Policy driver: Federal tax incentives and state procurement frameworks continue to support deployment, although tariff exposure and FEOC-related sourcing constraints may create temporary timing volatility in 2026 and 2027 rather than reversing the long-run trajectory.[213] [25]
- Market interpretation: The path from 43.6 GW to 100+ GW is best understood as a scale-up of an already mainstream grid asset class, not as an early-stage adoption forecast.[195] [205]

6.2

Global Deployment Records and the Battery Storage Supercycle

Global deployment set a new benchmark in 2025. The IEA reports 108 GW of new battery storage capacity deployed worldwide in 2025, up 40% year on year, while BloombergNEF reports 112 GW and 307 GWh of annual additions, marking the first time the sector exceeded 100 GW in a single year; the small difference reflects source methodology, but both confirm the same structural inflection.[19, 22]

The supercycle framing is justified because storage demand is now being pulled by several reinforcing forces at once. These include rapid solar and wind additions, falling battery prices, the need for short-cycle grid flexibility, rising electrification, and the emergence of new power-demand centers that value fast, modular capacity.[28, 194, 212]

- Record scale: Battery storage was the fastest-growing power technology in 2025, with annual additions exceeding the historical peak annual additions of natural gas capacity, a notable benchmark for investors assessing whether storage has become a core infrastructure category.[194, 197]
- Utility-scale dominance: Around 87 GW of global battery additions in 2025 came from utility-scale projects, or roughly four-fifths of the total, showing that the market is increasingly driven by grid applications rather than only distributed or behind-the-meter use cases.[28]
- China-led acceleration: China remained the anchor market, accounting for around 60% of global additions in IEA data and 54% in BloombergNEF data, with the United States the second-largest market and Europe following behind.[19, 22]
- Duration shift: The market is also maturing in design terms, with more systems built for multi-hour energy shifting rather than only frequency response, which broadens the addressable revenue stack and strengthens long-term deployment economics.[27]
- Supercycle implication: This is not merely a cyclical rebound from low prices. It is a structural expansion phase in which storage is becoming embedded in generation planning, transmission deferral, capacity procurement, and industrial load-serving strategies across multiple regions.[28, 212]

6.3

Pipeline, Planned Additions, and Regional Hotspots

The next phase of growth will be shaped less by whether storage expands and more by where it expands fastest. Current evidence points to a concentrated set of regional hotspots led by China, the United States, and selected high-renewables markets, but with a widening second tier of countries where storage is moving from pilot scale to system-planning relevance.[28, 22]

In the United States, Texas and California remain the most visible utility-scale hubs because they combine large solar pipelines, strong peak-shifting value, and market structures that reward flexible capacity. EIA highlights California's continued concentration of large projects, while Wood Mackenzie points to robust contracting activity and pipeline depth tied increasingly to large-load and data-center demand, especially in Texas and PJM-linked regions.[195, 211, 204]

Title — Key battery storage deployment hotspots through 2030

REGION	CURRENT SIGNAL	OUTLOOK IMPLICATION
China	Largest global market, with around 54% to 60% of 2025 additions depending on source.	Remains the dominant volume driver and cost anchor for global deployment.
United States	43.6 GW operational by end-2025, with utility-scale as the main growth engine.	Likely to exceed 100 GW utility-scale around 2030 if current build rates persist.
Europe	Growing from a smaller base, supported by flexibility needs and high-renewables integration.	Important second-tier growth region, but still behind China and the U.S. in scale.
Australia	New projects are increasingly designed for 2 hours or more, with average duration rising.	Remains a leading test bed for higher-duration merchant and grid-support storage.

Source: [19] [27] [22] [195]

- China: The largest hotspot by far, supported by manufacturing scale, grid-balancing needs, and continued utility-scale deployment momentum even after policy adjustments to co-location rules.[28] [22]
- United States: The deepest near-term pipeline is concentrated in solar-rich and fast-growing load regions, with California, Texas, and PJM-adjacent territories standing out for different reasons including resource adequacy, merchant spreads, and data-center demand.[195] [204]
- Europe: Growth is increasingly linked to flexibility needs in high-renewables systems and to the economics of shifting solar and wind output, though deployment remains more fragmented by market design and national policy than in China or the U.S.[28]
- Australia: A smaller market in absolute terms, but strategically important because project duration is lengthening and storage is being integrated more directly into system reliability planning.[27]
- Pipeline risk lens: The main constraints on converting pipeline into operating capacity are no longer technology readiness, but interconnection queues, trade-policy friction, local permitting, and revenue-stack evolution. For investors, that means regional hotspot analysis should focus as much on execution conditions as on headline demand growth.[205] [28]

7

Emerging Technologies and Long-Term Cost Trajectories

7

Emerging Technologies and Long-Term Cost Trajectories

Emerging storage technologies matter less because they displace lithium ion immediately and more because they can change the shape of future cost curves by altering chemistry risk, duration economics, and the balance between energy and power costs. For investors, the key analytical distinction is between technologies that can plausibly compress short duration battery costs within this decade, such as sodium ion, and technologies that are more likely to create option value in specific premium segments first, such as solid-state and several long-duration storage platforms.[93] [243] [11]

The practical implication for long-term cost modeling is that a single lithium ion learning curve is becoming less sufficient. Sodium ion introduces a credible low-cost branch for stationary storage, solid-state remains a higher-performance but still pre-scale pathway, and long-duration technologies such as flow batteries and gravity-based systems could flatten the cost penalty of moving from four-hour storage to multi-hour and multi-day flexibility if commercialization and manufacturing scale improve as expected.[245] [239] [231]

7.1

Sodium-ion Batteries: CATL Scaling, IRENA $40/kWh Projection

Sodium-ion has moved from a technology watchlist item to an investable commercialization signal in 2025 and 2026. CATL unveiled Naxtra in April 2025 as what it described as the world's first mass-produced sodium-ion battery, and by April to May 2026 CATL and HyperStrong had converted that platform into a three-year 60 GWh sodium-ion energy storage supply agreement, which CATL described as the official entry of sodium-ion energy storage into the GWh-scale deployment era.[96] [94] [93]

The cost case is strategically important because sodium-ion attacks one of the main residual risks in lithium-based cost curves, namely raw-material concentration and lithium price volatility. IRENA's 2025 sodium-ion technology brief, as reported by pv magazine, states that sodium-ion cell costs could fall to about USD 40/kWh, while CATL's commercialization push suggests the industry is now testing whether that theoretical cost advantage can survive the transition from pilot lines to bankable supply chains.[235] [93]

The main reason sodium-ion matters for stationary storage is not superior energy density, but a potentially lower and more resilient cost floor in applications where footprint is manageable and cycle life, safety, and cold-weather performance are more important than gravimetric density. Academic scenario work comparing more than 6,000 sodium-ion and lithium-ion pathways found that sodium-ion could reach competitiveness with low-cost lithium-ion variants in the early 2030s under favorable learning, supply-chain, and technology-improvement assumptions, which supports treating sodium-ion as a real medium-term branch in cost-curve analysis rather than a speculative outlier.[11]

- Commercial signal: The 60 GWh CATL HyperStrong agreement is the strongest current evidence that sodium-ion is moving beyond demonstration into scaled stationary procurement.[94] [93]
- Cost signal: The IRENA-linked USD 40/kWh figure is a cell-level projection, not a turnkey system benchmark, so analysts should not map it directly into installed project capex without BOS, PCS,

thermal, and EPC adders.[235, 239]

- Strategic use case: Sodium-ion is most likely to compete first in cost-sensitive stationary storage and selected commercial vehicles, not in premium high-energy-density segments.[96, 11]
- Investor implication: The technology should be modeled as an option that can compress future pack and cell costs and reduce lithium exposure, but not yet as the base-case chemistry for 2026 project underwriting.[93, 11]

7.2 Solid-State Storage Potential

Solid-state batteries remain one of the most important long-term technology options, but as of August 25, 2026 they are still better understood as a pre-commercial scaling story than as a near-term cost disruptor for mainstream stationary storage. Company disclosures from QuantumScape indicate that 2026 priorities remain demonstration of scalable production, line ramp, and continuous improvement in productivity, quality, and reliability, which is consistent with a sector still proving manufacturability rather than delivering broad grid-scale cost competitiveness.[232]

The core attraction of solid-state is performance, not immediate cost leadership. Replacing liquid electrolyte with solid or semi-solid architectures can improve safety and enable lithium-metal pathways with higher energy density, but those advantages are most valuable first in premium applications where energy density and safety justify higher initial costs, such as advanced mobility, aerospace-adjacent uses, and high-performance devices, rather than commodity stationary storage where LFP already sets a low-cost benchmark.[232, 225]

For stationary storage, the long-term relevance of solid-state is still substantial because it could eventually improve safety margins, usable energy density, and degradation performance in constrained sites. However, there is not yet strong public evidence that all-solid-state systems will undercut LFP or sodium-ion on cost in mainstream four-hour grid applications this decade, so the prudent analytical treatment is to model solid-state as a premium technology pathway with optionality beyond 2030 rather than as a 2026 to 2030 base-case cost curve.[245, 232]

- Technology status: Public disclosures still emphasize pilot-line execution and scalable production proof points, indicating that manufacturing readiness remains the gating issue.[232]
- Cost trajectory: Near-term solid-state cost curves are likely to be above incumbent LFP and potentially above future sodium-ion in stationary applications because early production volumes are low and process complexity remains high. This is an inference based on current commercialization stage and incumbent cost leadership.[232, 245]
- Best analytical role: Use solid-state in scenario analysis for premium, safety-constrained, or space-constrained storage applications, not as the central benchmark for utility-scale cost forecasting through 2030.[225, 232]
- Market conclusion: Solid-state remains transformative in potential, but not yet transformative in bankable stationary-storage cost data.[232, 243]

7.3

Long-Duration Storage: Flow Batteries, Gravity-Based Systems

Long-duration storage is where future cost trajectories may diverge most sharply from today's lithium-ion-dominated market. IRENA's 2026 analysis argues that multi-day and seasonal reliability challenges cannot be solved cost-effectively by short-duration lithium-ion alone, and identifies long-duration technologies such as flow batteries, compressed-air systems, pumped heat, thermal storage, and other non-lithium options as increasingly important complements as systems move toward higher renewable penetration.[243]

Flow batteries are the most analytically mature electrochemical LDES alternative because their architecture separates power and energy costs. That matters for cost curves because once the stack and balance-of-plant are installed, extending duration mainly requires more electrolyte and tank capacity, which can reduce unit energy cost as discharge duration rises. DOE's Storage Innovations 2030 flow battery assessment projected 2030 LCOS of about USD 0.15 to USD 0.16/kWh for 10-hour, 100 MW to 1,000 MW systems and framed further progress toward the USD 0.05/kWh Long Duration Storage Shot target as dependent on advances in active electrolytes, scalable manufacturing, controls, and system design.[233] [239]

Sandia's June 2026 industry update reinforces that flow batteries are moving from laboratory relevance toward practical deployment, but still face commercialization challenges in scaling from demonstrations to repeatable projects. That makes flow batteries credible for long-duration scenario modeling today, but still sensitive to execution risk, supply-chain maturity, and bankability assumptions.[231]

Gravity-based systems are earlier-stage and more site- and design-specific, but they remain relevant because they target the same strategic gap as other LDES technologies: lowering the cost of storing energy for many hours or days without relying on large quantities of lithium-based cells. IRENA's 2025 and 2026 materials classify gravity-based systems as emerging or under demonstration rather than broadly commercial, so they should be treated as optionality in long-range portfolios rather than as near-term benchmark technologies.[245] [243]

Title — Long duration storage technology outlook

TECHNOLOGY	CURRENT MARKET POSITION	COST TRAJECTORY IMPLICATION
Flow batteries	Commercial but still scaling in grid applications.	Better suited than lithium-ion to cost declines at longer durations because energy and power costs are partly decoupled.
Gravity-based systems	Demonstration to early commercial stage.	Potentially attractive for multi-hour to multi-day storage, but cost curves remain highly project-specific and immature.
Other LDES categories	Includes compressed-air, thermal, and pumped heat systems.	Broad evidence points to declining costs by 2030, but technology choice remains use-case dependent.
Lithium-ion for comparison	Dominant in short-duration storage today.	Remains strongest for daily cycling, but becomes less cost-optimal as duration extends into multi-day needs.

Source: [243] [245] [233]

- Flow battery advantage: Duration extension is structurally cheaper than in lithium-ion because additional energy capacity does not require proportional duplication of the full power train.[233]
- Commercialization constraint: The main barriers are not only chemistry performance, but manufacturing scale, project standardization, and bankability.[231] [239]
- Gravity-system role: Best treated as a niche but potentially valuable mechanical-storage option where geography, permitting, and project design align, not as a universal substitute for batteries.[245]
- Portfolio implication: The most credible long-term cost trajectory is not one winner replacing all others, but a layered market in which lithium-ion dominates daily cycling, sodium-ion pressures low-cost stationary segments, and LDES technologies capture the duration ranges where lithium-ion economics weaken.[243] [239]

8

Practical Guidance for Analysts and Decision-Makers

8

Practical Guidance for Analysts and Decision-Makers

This section translates the report's earlier findings into a decision workflow for investors, developers, utilities, and policy analysts. The central principle is simple: do not ask one cost curve to answer every question. A pack-price series is useful for tracking manufacturing and procurement direction, while a duration-specific installed-cost curve is more appropriate for project underwriting, resource planning, and policy design.[50, 183, 20]

For practical use, analysts should treat battery cost curves as layered evidence rather than competing truths. The most robust approach is to anchor the present with observed market data, normalize all sources to a common boundary and duration, and then extend forward with transparent scenario assumptions on learning, policy, and regional frictions.[183, 28, 3]

8.1

A Framework for Selecting and Interpreting Cost Curves

A useful selection framework starts with the decision being made, not with the source brand. NREL's ATB and 2025 utility-scale update are designed for transparent, duration-specific installed-cost analysis, while IEA and Bloomberg-derived pack benchmarks are better for identifying current market direction and regional price dispersion rather than turnkey project capex.[50, 183, 20]

- Step one: Define the decision boundary. Use cell or pack curves for manufacturing competitiveness, supplier positioning, and chemistry tracking; use full installed system curves for project finance, utility procurement, and policy cost-effectiveness tests.[50, 3]
- Step two: Normalize the denominator. Convert all sources to a common basis such as usable $/kWh, nameplate $/kWh, or installed $/kW before comparing them, because NREL explicitly models storage across 2, 4, 6, 8, and 10 hour durations and the apparent cost ranking changes with duration.[50]
- Step three: Identify methodology. Bottom-up engineering curves are stronger for scenario consistency, while empirical market curves are stronger for current procurement reality; hybrid use is usually best when the decision spans both near-term execution and long-term planning.[183, 3]
- Step four: Check whether degradation and augmentation are included. Life-cycle studies show that omitting degradation can overstate profitability and distort the effective cost of delivered storage services.[2, 41]
- Step five: Separate endogenous learning from exogenous overlays. Learning-driven declines should be distinguished from tariffs, local-content rules, interconnection costs, and regional labor effects, because those factors can flatten installed-cost curves even when pack prices continue to fall.[20, 28]
- Step six: Match the curve horizon to the use case. For bids, procurement, and investment committee timing, start with current market evidence; for 2030 pathway work, start with a transparent modeled curve and then sanity-check it against current market benchmarks.[183, 20]

Title — Decision use cases and preferred curve types

USE CASE	PREFERRED CURVE TYPE	WHY IT FITS
Supplier benchmarking	Cell or pack price curve	Best for chemistry, manufacturing scale, and procurement frontier tracking.
Project finance	Full installed system curve	Captures PCS, BOS, EPC, interconnection, and soft costs.
Resource planning	Duration-specific modeled installed-cost curve	Preserves comparability across 2 to 10 hour systems and scenarios.
Policy analysis	Regional installed-cost curve with scenario overlays	Reflects tariffs, localization, and non-cell cost frictions.
Market timing	Empirical current-price curve	Better captures oversupply, discounting, and short-term procurement windows.

Source: [50] [183] [20]

8.2

Red Flags When Reading Published Projections

Published projections usually become misleading not because the numbers are fabricated, but because the scope and assumptions are hidden or easy to misread. The most common failure is treating a low pack-price headline as if it were equivalent to a financeable installed project cost, even though NREL's framework explicitly separates battery pack cost from inverter, BOS, and installation layers.[50] [183]

- Boundary ambiguity: Be cautious when a source says "battery cost" without specifying whether it means cell, pack, system equipment, or full installed project cost.[50]
- Duration blindness: Treat any $/kWh or $/kW figure with caution if the discharge duration is not stated, because duration materially changes comparability across systems.[50]
- Geography-free forecasts: Discount projections that present a single global curve as decision-ready for the United States, Europe, and China alike, given the IEA's evidence of widening regional price divergence and supply-chain dependence.[20]
- Missing policy assumptions: Treat projections as incomplete if they do not disclose tariff, trade, localization, or subsidy assumptions, because these can materially alter delivered cost even when underlying battery manufacturing costs fall.[28]
- Over-smooth trajectories: Be skeptical of curves that decline in a perfectly steady line. Real markets are affected by oversupply, commodity rebounds, and policy shocks, so smooth curves may be useful for planning but weak for procurement timing.[183] [20]
- No degradation treatment: If a projection is used for revenue or IRR analysis but excludes degradation, augmentation, or replacement timing, it likely overstates economic performance.[2] [41]
- Chemistry generalization: Be cautious when a source extrapolates one lithium-ion curve across all stationary applications without clarifying LFP versus NMC or the possibility of sodium-ion entry in later years.[20] [3]
- Hidden base year issues: Some projections are anchored to older base years and then extrapolated forward, which can miss fast market resets. NREL itself notes the importance of base-year assumptions and backward extrapolation conventions in ATB methodology.[265] [183]

A practical screening rule is to reject any projection that cannot answer five questions in one page: what boundary is measured, what duration is assumed, what geography is represented, what methodology is used, and which policy or commodity assumptions are embedded. If any of those are missing, the curve may still be directionally useful, but it is not decision-grade evidence.[183, 20]

8.3

Reconciling Disagreements Across Sources

When sources disagree, the first task is not to choose a winner but to diagnose the source of divergence. In most cases, disagreement can be decomposed into four buckets: boundary mismatch, duration mismatch, methodology mismatch, and scenario mismatch.[50, 183, 3]

- Normalize first: Convert each source to a common year, currency, geography, duration, and cost boundary before interpreting the spread. Many apparent contradictions disappear once pack-level and installed-cost series are separated.[50, 20]
- Build a bridge model: Use a simple reconciliation stack that starts with pack cost and adds PCS, thermal, fire safety, BOS, EPC, interconnection, and soft-cost assumptions to estimate a comparable installed-cost range. This is effectively the logic embedded in NREL’s bottom-up structure.[50, 183]
- Use empirical data for the base year: Anchor 2025 to 2026 starting values to observed market evidence, then apply modeled forward trajectories. This reduces the risk of carrying stale base-year assumptions into long-run forecasts.[20, 28]
- Reconcile by scenario, not by average: If one source assumes benign trade conditions and another assumes tariff friction or localization costs, preserve both as explicit scenarios rather than averaging them into a false midpoint.[20, 28]
- Distinguish trend disagreement from level disagreement: Two sources may agree on the long-run direction but disagree on the starting point, or vice versa. That distinction matters because procurement decisions are more sensitive to level, while strategic planning is more sensitive to slope.[183, 3]
- Carry ranges into valuation: For investment work, use low, base, and high cases tied to explicit assumptions on pack prices, BOS compression, tariffs, and degradation rather than a single deterministic curve.[2, 20]
- Document irreducible uncertainty: If after normalization the spread remains wide, preserve it. A wide range may be the correct analytical output in a market shaped by policy volatility, regional divergence, and changing chemistry mix.[20, 28]

For decision-makers, the most defensible practice is to maintain a two-layer view of the market: a current-market layer based on observed pack and procurement evidence, and a planning layer based on normalized installed-cost scenarios. That approach preserves realism for near-term execution while keeping long-term strategy internally consistent.[183, 50, 20]

9

Conclusions and Forward Outlook

9

Conclusions and Forward Outlook

The study's central conclusion is that battery storage cost curves are still falling, but the market is no longer well described by a single headline number. The most important analytical shift for 2026 is from asking whether batteries are getting cheaper to asking which boundary, region, duration, and policy regime a given cost curve actually represents, because pack-level declines and installed-project economics are diverging more visibly as non-cell costs and regional frictions become more important.[20] [21] [28]

The forward outlook remains constructive. LFP continues to anchor the mainstream stationary market, global deployment has entered the 100 GW annual-additions era, and batteries are taking on a larger role in energy shifting and system flexibility, but the slope of realized regional cost curves will depend increasingly on tariffs, localization, interconnection, merchant revenue quality, and the pace at which sodium-ion and long-duration alternatives become commercially credible complements rather than laboratory options.[19] [28] [11]

9.1

Key Takeaways on Cost Trajectory and Market Direction

- Cost trajectory: The broad direction remains downward, but the decline is uneven across boundaries. BloombergNEF reported a global average lithium-ion pack price of $108/kWh in 2025 and a stationary-storage pack price of $70/kWh, while the IEA and NREL-style evidence used throughout this report indicates that full installed system costs fall more slowly because PCS, BOS, EPC, interconnection, and soft costs do not compress at the same rate as cells and packs.[21] [20]
- Market structure: The market is consolidating around LFP as the default chemistry for mainstream stationary storage. The IEA states that LFP accounted for over 90% of global stationary battery storage installations in 2025, which means most near-term cost curves should be interpreted as LFP-led unless a source explicitly models a chemistry transition.[20]
- Regional direction: Global averages are becoming less decision-useful on their own. China remains the cost frontier, while North America and Europe continue to carry meaningful premiums, so future installed-cost convergence is likely to be slower than pack-price convergence unless trade frictions ease and local manufacturing plus BOS productivity improve.[21] [19]
- Demand outlook: Deployment momentum is now strong enough that batteries should be treated as core grid infrastructure rather than an emerging niche. The IEA reports that battery storage was the fastest-growing power technology in 2025, while its 2026 commentary shows batteries increasingly designed for energy shifting and multi-service system roles, supporting continued volume growth through 2030 even if annual pricing becomes more volatile.[19] [28]
- Technology outlook: The most credible medium-term market shape is layered rather than winner-take-all. Lithium-ion, especially LFP, should remain dominant for daily cycling, sodium-ion is emerging as a plausible low-cost branch for selected stationary applications in the early 2030s, and long-duration technologies are more likely to complement than displace lithium-ion in higher-duration use cases.[11] [28] [3]

9.2
Remaining Uncertainties and Risks

- Trade and policy risk: The largest near-term uncertainty is no longer whether manufacturing costs can fall, but whether those declines reach end markets. Tariffs, local-content rules, export controls, and industrial-policy responses can flatten regional installed-cost curves even when global pack benchmarks continue to improve.[28] [19]
- Non-cell cost stickiness: A major risk to overly bullish cost forecasts is that non-battery hardware and project execution costs become the dominant source of variance. If interconnection, labor, fire-safety systems, power electronics, and EPC overhead remain sticky, future pack-price declines will translate only partially into lower turnkey capex.[20] [3]
- Commodity volatility: Lithium remains the most direct raw-material risk for stationary storage, even in an LFP-dominant market. The broader lesson from recent market behavior and scenario literature is that commodity shocks are likely to create step changes and temporary reversals rather than a smooth learning curve, which argues against deterministic single-line forecasts.[20] [11]
- Revenue-stack uncertainty: Falling capex does not guarantee stronger project returns. Recent academic work on battery bidding and life-cycle economics shows that price uncertainty, degradation, and operating strategy can materially change realized profitability, especially as more batteries compete for similar arbitrage and ancillary-service revenues.[277] [2]
- Technology transition risk: Sodium-ion, flow batteries, and other alternatives are strategically important, but their commercialization timelines remain uncertain. The risk is two-sided: analysts may either ignore credible substitution pathways and overstate lithium dependence, or over-credit emerging technologies before bankable supply chains, performance data, and repeatable project execution exist.[11] [28]
- Forecasting risk: Published curves will continue to disagree because the market itself is becoming more segmented by duration, geography, use case, and policy regime. That disagreement should be treated as a signal of structural complexity rather than as evidence that one source is necessarily wrong.[3] [2]

9.3
Recommendations by Stakeholder Type

- Investors: Underwrite to full installed cost and life-cycle economics, not pack-price headlines. Use empirical market data to anchor the base year, apply explicit low, base, and high cases for tariffs, BOS compression, and degradation, and test whether lower capex is being offset by weaker merchant spreads or ancillary-service saturation.[21] [2] [277]
- Policymakers: Design support frameworks around delivered system economics rather than global manufacturing averages. Policies that reduce interconnection delays, permitting friction, and BOS soft costs may improve deployment economics as much as policies focused only on cell manufacturing, while trade and localization measures should be evaluated for both resilience benefits and near-term cost pass-through.[19] [28]
- Utilities and system planners: Use duration-specific installed-cost curves and model batteries as flexibility assets with multiple system roles. Planning assumptions should reflect the market shift toward energy shifting, four-hour-plus configurations, and co-optimization with renewables, transmission constraints, and demand growth rather than relying on legacy short-duration ancillary-service archetypes.[28] [115]

- Developers and industry leaders: Prioritize execution discipline over headline equipment savings. Competitive advantage will increasingly come from supply-chain optionality, standardized system design, faster interconnection and permitting, and the ability to manage non-cell cost inflation, not only from securing lower battery prices.[20] [21]
- Corporate strategists and manufacturers: Treat LFP as the base case, but maintain option value in sodium-ion and selected long-duration platforms. The most prudent strategy is not to bet on immediate displacement of lithium-ion, but to build portfolios that can respond if sodium-ion learning accelerates or if long-duration markets begin rewarding technologies with structurally better economics beyond four hours.[11] [3]
- Analysts and research teams: Maintain a two-layer framework. Track current market reality with observed pack and procurement data, then translate that into normalized installed-cost scenarios with explicit assumptions on duration, degradation, policy, and geography, preserving uncertainty ranges rather than forcing false precision.[21] [20] [2]

10

References and Data Sources

10

References and Data Sources

This section consolidates the principal references and data inputs used across the study, grouped by evidentiary role rather than by appearance in the text. Citations are presented in a consistent professional format to distinguish primary market data, institutional modeling sources, academic research, and supporting commentary current to August 25, 2026.

10.1

Primary Sources

- International Energy Agency. *Global Energy Review 2026: Technology, Battery Storage*. Paris: IEA, 2026. Used for global battery storage additions, regional deployment shares, and the characterization of battery storage as the fastest-growing power technology in 2025.[19] [292]
- International Energy Agency. *Electricity 2026: Flexibility*. Paris: IEA, 2026. Used for utility-scale deployment trends, duration shifts, and the expanding system role of batteries in power-sector flexibility.[27] [295]
- International Energy Agency. *Global EV Outlook 2026*, section on electric-vehicle batteries. Paris: IEA, 2026. Used for regional battery price comparisons, LFP cost advantage versus NMC, and LFP's dominant share in stationary storage.[20]
- International Energy Agency. *Energy Technology Perspectives 2026*. Paris: IEA, 2026. Used for trade, manufacturing, and delivered-cost effects associated with tariffs, localization, and battery supply-chain restructuring.[297]
- International Energy Agency. *Global Critical Minerals Outlook 2026*. Paris: IEA, 2026. Used for lithium and cobalt price movements, supply concentration, export-control risk, and commodity-volatility context.[123]
- BloombergNEF. *Lithium-Ion Battery Pack Prices Fall to $108 Per Kilowatt-Hour, Despite Rising Metal Prices*. New York: BloombergNEF, December 9, 2025. Used for the 2025 global average pack price of USD 108/kWh, China at USD 84/kWh, regional premiums in North America and Europe, and stationary-storage pack pricing at USD 70/kWh.[21]
- National Renewable Energy Laboratory. Cole, W. et al. *Cost Projections for Utility-Scale Battery Storage: 2025 Update*. Golden, CO: NREL, 2025. Used for bottom-up installed-cost trajectories, duration-specific cost treatment, and scenario framing in the source-comparison sections.[183] [48]
- National Renewable Energy Laboratory. *Annual Technology Baseline 2024b: Utility-Scale Battery Storage*. Golden, CO: NREL, 2024. Used for cost-boundary definitions, pack-plus-BOS structure, and duration normalization across 2-hour to 10-hour systems.[50]
- U.S. Energy Information Administration. *Battery storage capacity in the United States is expected to nearly double by the end of 2026*. Washington, DC: EIA, 2025. Used for U.S. operating-capacity benchmarks and near-term utility-scale additions.[195]
- U.S. Energy Information Administration. *Short-Term Energy Outlook*, January 2025 archive. Washington, DC: EIA, 2025. Used for projected U.S. battery additions in 2025 and 2026.[201]
- Wood Mackenzie and American Clean Power Association. *U.S. Energy Storage Monitor* and associated 2025 to 2026 releases. Used for quarterly deployment data, five-year U.S. outlooks, tariff sensitivity, and market pipeline commentary.[291] [293] [294]

- Wood Mackenzie. *U.S. energy storage market sets Q1 2026 records across sectors*. Used for cumulative U.S. outlooks toward 2031 and utility-sector share assumptions.[205]
- S&P Global Commodity Insights. *Declining costs, shifting revenues, evolving business case for battery storage*. Used for revenue-aware interpretation of storage economics and the distinction between falling capex and changing merchant value.[174]
- S&P Global Commodity Insights. *US lithium-ion battery imports rebound in Q2 despite trade friction*. Used for recent U.S. import-volume evidence and trade-friction effects as of August 24, 2026.[127]
- CATL. *Naxtra* product announcement and subsequent sodium-ion storage partnership releases with HyperStrong, 2025 to 2026. Used for commercialization evidence on sodium-ion scaling and GWh-scale deployment claims.[96] [94] [93]
- U.S. Department of Energy. *Achieving the Promise of Low-Cost Long Duration Energy Storage*. Used for long-duration storage cost targets and technology pathway framing.[239]
- U.S. Department of Energy. *Technology Strategy Assessment: Flow Batteries*. Used for flow-battery LCOS benchmarks and duration-specific cost logic.[233]
- International Renewable Energy Agency. *24/7 Renewables: Power System Reliability and Cost in 2026* and related 2025 technology materials. Used for long-duration storage positioning, gravity-based storage status, and sodium-ion cost-potential references cited in the emerging-technology discussion.[243] [245]
- Academic primary research used directly in the analytical framework included: Ziegler and Trancik, *Re-examining rates of lithium-ion battery technology improvement and cost decline*; Yao, Benson, and Chueh, *How quickly can sodium-ion learn?*; Yang et al., *Life cycle economic viability analysis of battery storage in electricity market*; Khalilisenobari and Wu, *Impact of Battery Degradation on Market Participation of Utility-Scale Batteries*; and Bastianin, Li, and Shamsudin, *Forecasting the Volatility of Energy Transition Metals*. These papers were used for learning-rate interpretation, chemistry-transition scenarios, degradation-aware economics, and commodity-volatility framing.[3] [11] [2] [41] [116]

10.2
Secondary Sources

- Secondary sources were used selectively to contextualize, triangulate, or interpret primary market and institutional data rather than to establish core numerical claims.
- IEA commentaries supported interpretation of batteries' expanding system role, the shift toward energy shifting, and sodium-ion commercialization momentum. These were used as explanatory supplements to the main IEA datasets and reports.[28] [98] [156]
- Wood Mackenzie press releases and market commentaries were used as secondary commercial intelligence for tariff impacts, large-load demand implications, and the framing of a storage supercycle. These sources informed market interpretation but were not treated as substitutes for underlying deployment datasets.[25] [125] [212] [204]
- Public-facing ACP summaries of the U.S. Energy Storage Monitor were used as secondary access points to Wood Mackenzie and ACP findings where the full subscription report was not directly available.[293] [294]
- Public regulatory and presentation materials were used as secondary corroboration for proprietary-source references, especially where BloombergNEF or S&P Global findings were

reproduced in public proceedings or stakeholder decks.[30, 177]

- Sandia National Laboratories' 2026 industry update on flow batteries was used as a secondary source for commercialization status and deployment challenges in long-duration electrochemical storage.[231]
- Company disclosures were used as secondary evidence for technology-readiness assessment where broad public market data were limited, particularly for solid-state batteries. QuantumScape investor and SEC materials were used in this way.[232, 106]
- Trade and technology journalism was used sparingly and only where it pointed back to institutional or company-originated claims, such as the reporting of IRENA-linked sodium-ion cost projections.[235]
- Methodological note: where a secondary source summarized proprietary research, the report prioritized the original publisher when a direct public page was available. Secondary sources were therefore used mainly for interpretation, public accessibility, or corroboration, not as the sole basis for key statistics.

10.3

Data Sets Used

- IEA global battery storage deployment data: Used for annual additions, regional shares, and utility-scale versus distributed deployment structure. Source base included the IEA's *Global Energy Review 2026* and *Electricity 2026* battery-storage datasets.[19, 27]
- IEA battery price and chemistry dataset: Used for regional battery-price comparisons, LFP versus NMC cost differentials, and stationary-storage chemistry shares.[20]
- BloombergNEF Battery Price Survey 2025: Used for global average pack prices, China benchmark pricing, regional premiums, and stationary-storage pack pricing. This is a proprietary market dataset accessed through BloombergNEF's public summary page and corroborating public reproductions.[21, 30]
- NREL utility-scale battery cost model inputs: Used for installed-cost decomposition, duration normalization, and scenario-based cost projections. These data are bottom-up engineering estimates rather than transaction-price observations.[50, 183]
- U.S. EIA utility-scale battery capacity data: Used for U.S. operating-capacity benchmarks and near-term additions. These data are public and administrative in nature, derived from EIA reporting and outlook products.[195, 201]
- Wood Mackenzie and ACP U.S. Energy Storage Monitor dataset: Used for quarterly U.S. deployment tracking, five-year outlooks, and market segmentation. This is a proprietary commercial dataset referenced through public summaries and release pages.[291, 293]
- S&P Global trade-flow and market-economics data: Used for U.S. battery import volumes and for revenue-aware interpretation of storage economics. This is a proprietary commercial dataset accessed through public reporting pages.[127, 174]
- IEA critical-minerals market data: Used for lithium, cobalt, and broader energy-transition-metal price and concentration analysis.[123]
- Academic scenario and modeling datasets embedded in cited papers: Used qualitatively rather than re-estimated directly. These included harmonized lithium-ion historical price and performance series, sodium-ion versus lithium-ion scenario sets exceeding 6,000 modeled pathways, and degradation-aware battery market-participation case-study data.[3, 11, 2, 41]

- Proprietary versus public data treatment: Public institutional datasets from IEA, NREL, DOE, EIA, and IRENA formed the report's reproducible backbone. Proprietary datasets from BloombergNEF, Wood Mackenzie, and S&P Global were used where they materially improved current-market accuracy, especially for 2025 to 2026 pricing, deployment monitoring, and trade-sensitive market intelligence.

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